University of Nottingham
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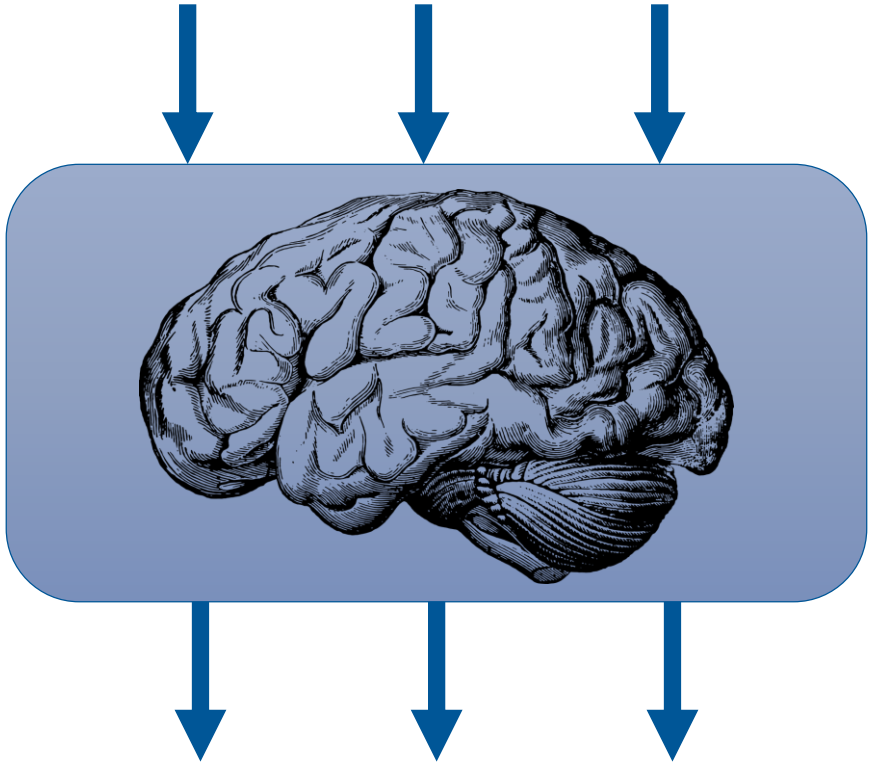
Designing Intelligent Agents: Coursework Introduction

Coursework Overview

- The module coursework is a project on a topic of your choice within the module, which must include the following components:
  - One more *autonomous intelligent agents*
  - …acting in some kind of *environment*
  - …about which you ask a *scientific or technical question* (or, set of related questions)
  - …and which you evaluate using *a set of structured, rigorous experiments*
- This is then presented in a report.
- In addition, Each MSc student needs to do a (3-5)-minute presentation to explain what you do

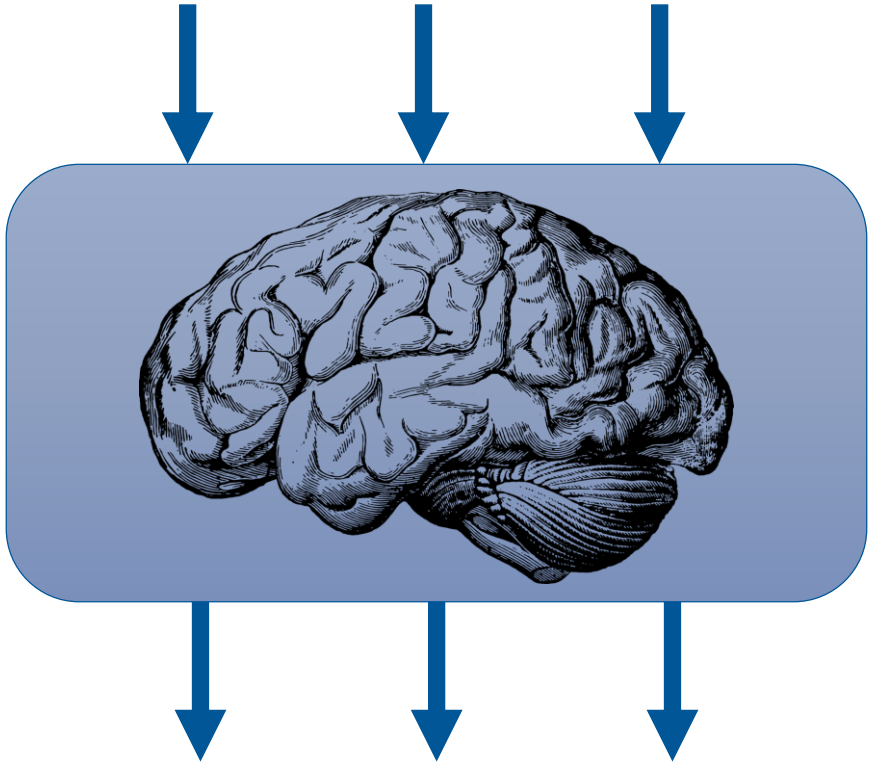
Autonomous Intelligent Agent

- An autonomous intelligent agent is a computer system that takes in information, uses some kind of AI to process it, and then selects some kind of action that changes the environment
- Often, there is:
  - a loop by which this change to the environment prompts the next set of inputs
  - multiple interacting agents giving rise to emergent behaviour
- So, not just something that learns into something static like a set of labels



Autonomous Intelligent Agent

- Intelligence could be anything from a reactive system or state machine that doesn't learn anything (but perhaps produces interesting emergent behaviour when placed in an environment with other agents), to a learning system using some AI/ML algorithm
- No preference for what kind of intelligent system you use—it is the quality of your understanding and how you analyse it that matters



Specification: Environment

- **An Environment.** This is the (virtual) place where the agents will operate. It could be one of:
  - A simulation of a physical environment in which mobile robotic agents move. This could be the simulation used in the practical in the semester (perhaps extended), a robot environment such as *The Player Project,* or a project in Unity or a similar game environment if you are familiar with one from elsewhere.
  - A language environment, for example where humans are interacting using written or spoken language with one or more intelligent agents,
  - The *Bristol Stock Exchange* system or a similar simulation of some aspect of the economy or society
  - A game environment such as *Ms. PacMan* , the *Open Racing Car Simulator,* RoboCup or similar
  - One of the more complex task environments from the *OpenAI Gym*
  - Please refer to the coursework description
- There is no need to develop the environment yourself—the focus of the project will be on the agents in the environment (robots, trading agents, game-playing agents, autonomous drivers, language agents, etc.) – but it is likely that you will set up the details of the environment to address your specific question. You are allowed to use the code from the classes, but please try to make it clear broadly which parts of the code are taken from the class examples, and which is your own work (we appreciate that this is sometimes complicated to do at a line-by-line level, but you should indicate this in broad terms).

Specification: Autonomous Agents

- You should introduce one or more autonomous intelligent agents into the environment, which use some kind of AI to solve a task.
  - Examples of AI could be an AI planning system such as Goal Oriented Action Planning, a search algorithm such as A* search, a genetic or swarm search, a reinforcement learning algorithm, fuzzy logic, or a hard-coded reactive or state-machine AI, machine learning…
  - The task will be one relevant to the environment: e.g. a robot vacuum cleaner clearing up dirt, a poet trying to write verse about the day's news, a trader trying to optimise its returns, a game player trying to get a high score in a game, etc.
- Language: You can use any language to do this. If you are planning to use anything other than Python, Java, C/C++, MATLAB, R, JavaScript, and mainstream web technologies such as HTML/CSS/JS, then please ask for approval.

Specification: Questions

- You should ask at least three specific questions (or a set of related questions) about your system. For example:
  - How do different approaches (a genetic algorithm, an A* search algorithm, a hard-coded heuristic) compare in terms of task performance?
  - How does the performance of the system change as we vary the number of agents in it?
  - If the system is trained on one version of the environment, does that learning transfer over to a new version of the environment
  - How do different kinds of communication/coordination between agents effect the efficiency of those agents on the task?
  - How much improvement does storing some information (e.g. a map of the environment) make compared to carrying out the task in a purely reactive way?
  - How do different kinds of sensing/perception systems affect the capacity of the agent to carry out its task?
  - How sensitive is the agent to error/noise?

Specification: Experiments

- You should answer your question by carrying out a set of experiments.
  - implement code that carries out a run of the agent's behaviour and measures performance
  - then, run that code multiple times to get a measure of average performance
  - then, repeat that process for the different conditions in your question, and use descriptive statistics, charts/visualisation, and/or inferential statistics (e.g. significance tests) etc to test your question
- Then, you should discuss the question using these experimental results as your evidence.
- If this evaluation involves asking people to interact with your code, then please read the notes on research ethics that are in the *Coursework* section of the module Moodle page.

Constraints

- Your project is expected to demonstrate around 90 hours of effort. So, you should not just run a basic reinforcement learning or convolutional neural networks library/tutorial on a simple environment from AI Gym or similar, though you can use these as part of your work.
- You are not allowed to develop a basic chatbot i.e. a single system that interacts with a human using intent matching, identity management, transaction processing, question answering etc. If you want to do something on language agents, you should do something that goes beyond this (e.g. a creative language system, a system that involves interactions between multiple language agents, or a system that involves a language-based interface to another system).
- You are free to use any readily available libraries, frameworks, APIs, services, datasets etc. as you want—but your work must demonstrate substantial work in using, combining and building on what you use. You are not permitted to simply re-submit coursework from another module, but you could potentially build on code that you have written previously and build on it in a new way. If your work extends or re-uses coursework from other modules, it must make a substantial and distinctive contribution beyond what was in those courseworks, and you must clearly identify what the new contribution is.

Examples

- Here are a few examples of things that you could do. You don't have to do one of these—indeed, we would prefer you to come up with your own idea—but, these would all be acceptable project ideas if you want to do them:
  - To take the "robot vacuum cleaner" from the practical, and experiment with different numbers of robots, and different coordination strategies (e.g. robots try to stay a fixed distance from each other, compared to sharing a map that they build up)
  - Explore how multiple language agents, each of which has a particular set of knowledge or perspective can, discuss/argue with each other to explore a topic.
  - Take a number of different trading strategies and run them in the *Bristol Stock Exchange* system with varying amounts of noise/uncertainty, to see how robust each strategy is.
  - Develop a game-playing agent or a control agent for an NPC or opponent in a game, and explore different strategies for playing (or learning to play) a game.
  - Take the "avoid the cats" problem from the class, and compare a number of strategies for the problem: warning the cats vs. moving out of the way, and learning when to act based on a simple statistical approach vs. a decision-tree approach.
  - Consider the problem of planning a robot's movement around a mapped environment (e.g. the map generated from WiFi triangulation introduced in one of the classes). Contrast A* search and genetic algorithms on this problem, and compare them both against random wandering.
  - Explore a swarm simulation such as the *Boids* flocking algorithm explored in one of the classes, and map out how the overall behaviour of the birds relates to the parameters in the algorithm.

Submission: Deadlines

- By 11:59 am on May 11, 2026 you should submit your work via Moodle. This deadline may be extended if you have a support plan or extenuating circumstances. Late submissions will incur a penalty of 5% per working day, in line with the standard University late policy.

Group

- Each Group has 3-5 members:
  - Enter your group information on the Module
  - The group leader (Member 1) should submit the report, code, and other materials.
  - **Each member should prepare one question and implement related experiments to answer it. (report and pre)**
  - Undergraduate and MSc students are not allowed to mix up.

Submission: Report

- A report, between 4000-8000 words (Based on the group size)
- The core ideas of your project:
  - A review of relevant ideas, technologies and research papers
  - How you designed the environment and agents
  - Technologies used, and challenges that you met in doing the implementation
  - Clearly state the question that you are trying to answer
  - How you set up and ran your experiments
  - The results from your experiments
  - A discussion of the questions in light of the experimental results
  - A conclusion, where you summaries the work, reflect on its successes and limitations, and briefly mention some ideas for how you would take the work forward if you had more time
  - Clarify each member’s contribution(one paragraph, what you do, the percentage of your workload)

Submission: Supplementary Material

- A copy of your code
- Anything else that you think would be helpful for the markers, e.g. sample outputs from your system, a link to a brief video demonstrating it working, reports from user studies/interviews etc.
- For undergraduate students, the mark for this portfolio of work will count as 100% of the module mark—there is no exam.

Presentation

- In addition to the portfolio of work, each MSc group should also do a 15-minute presentation about your work (**Date is May 15th 2026**). This should give an overview of your project and explain how it is informed by research ideas from AI and intelligent agents.
- For MSc students, the portfolio of work (including the report, code or other submission materials) will count for 90% of the marks on the module and the presentation for 10%—there is no exam.

Presentation

- Suggested structure
  - Intro to the environment and agents (1-2 slides)
  - Each student (3-4 slides):
    - Question you are asking
    - What experiments have you done
    - What are the results
    - What can be concluded from those results

- Q&A

Academic Integrity

- You should make any direct quotations clear both by using quotation marks and by providing a clear reference to the paper immediately after the quotation. If you are building on someone else's code (e.g. our code from the classes, open-source projects, etc.), please make it clear which aspects of the code are your work through the use of comments. The University has detailed advice about academic integrity, and submissions that demonstrate a lack of that integrity will be treated under appropriate disciplinary procedures.

How this will be assessed

- The intrinsic complexity of the overall project
- Background research and how you have used it to contextualise your work
- The choice of task environment and how you have used it/adapted it for your specific project
- The effective use of artificial intelligence and agent-based systems ideas from the course and your wider studies in designing your autonomous agents
- How clear your question(s) are, how well the experiments have been designed to answer them, and your level of rigour in planning and analysing the experiments
- How well the report answers the question by using the evidence from the experiments
- The overall clarity and structure of the report, appropriate use of scientific and technical English, and the quality of charts, diagrams, pseudocode where relevant
- The quality of reflection on the successes and limitations of the work
- (For students doing a presentation) the structure of the presentation, the clarity of explanations, and good use of slides or other visual aids

Q&A for coursework

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